Report 1: Project Description Report

Course: UML Class Diagram Modelling Assignment
Institution: Kigali Independent University (ULK)
Project Title: Hotel Booking Management System
Group Members:
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1. Project Title & Business Idea

Project Title: Hotel Booking Management System

Business Idea: A centralized digital software system that enables users to browse hotel room offerings, check availability, make reservations, and record payment transactions efficiently.

2. Problem Statement & System Purpose

Problem Statement: Traditional, paper-based, or manual hotel management leads to double bookings, inefficient room status updates, poor customer tracking, and delayed billing.

Purpose: To provide a reliable UML class model that structures data effectively for room availability, reservation management, and transactional record-keeping.

3. Project Scope

Included (In-Scope):

Managing guest profile data.
Tracking room availability, room types, and price calculations.
Creating, confirming, and canceling reservations.
Processing payment entries and generating payment receipts.

Excluded (Out-of-Scope):

Automated flight or transport service bookings.
Direct integration with hardware hardware locks or keycards.
Live external payment banking gateways.

4. Main Classes & Responsibilities

1.Guest: Stores personal details of customers and enables them to place reservations.
2.Room: Holds attributes regarding room numbers, types, rates, and real-time availability.
3.Reservation: Handles check-in/check-out dates, computes total cost, and maintains status.
4.Payment: Records details of monetary transactions attached to bookings.

5. System Requirements Brief

The system must allow guest identification.
Room records must be updated when a reservation is confirmed.
Every reservation must calculate total costs based on dates and room daily rates.
A payment record must be linked directly to a confirmed reservation.